

Introduction: Transformers. Large Language Models

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Agenda

- Introduction to NLP (evolution, core principles)
- Foundational Principles of NLP
- What are Vector Embeddings?
Understanding the Foundation of Data Representation
- Text cleaning pipeline
- Hands-on: Text cleaning
- Hands-on: Named Entity Recognition
- Introduction to Large Language Models (LLMs)
- Transformers Architecture

Intro to LLMs. Transformers



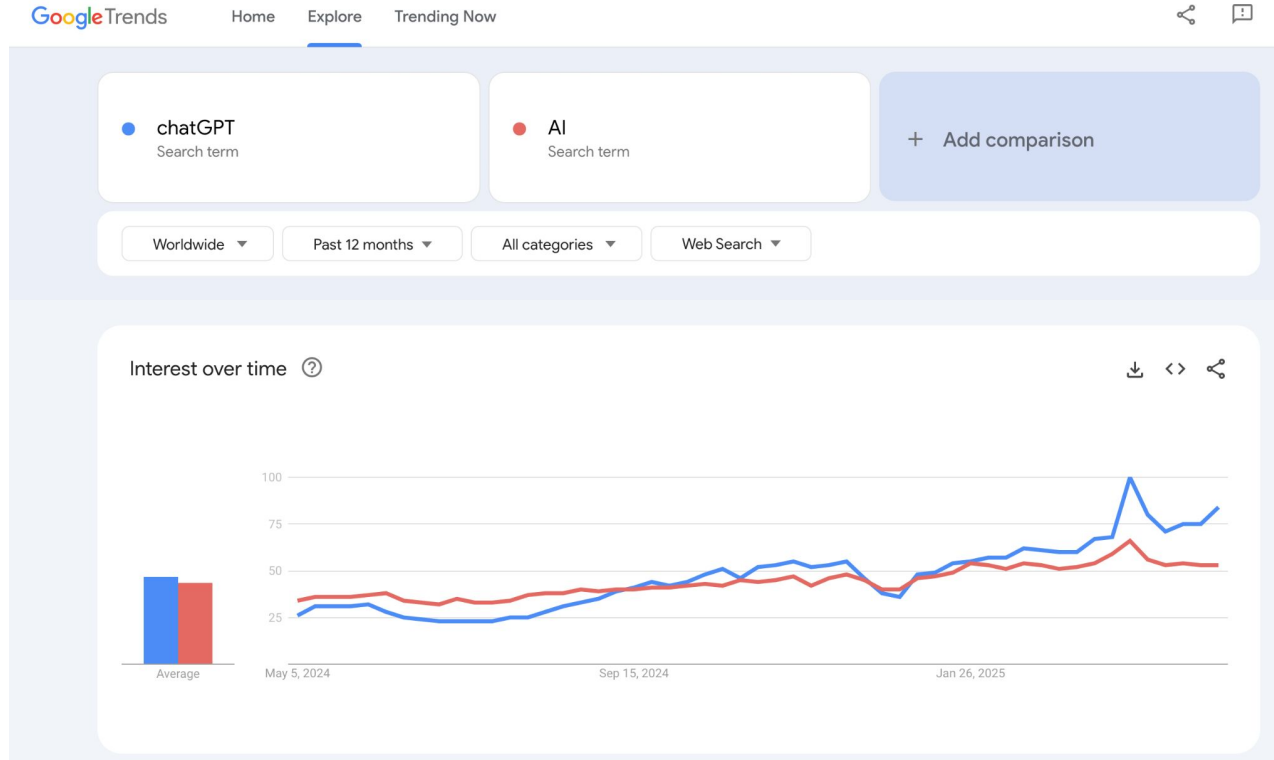
Andrej Karpathy ✓

@karpathy

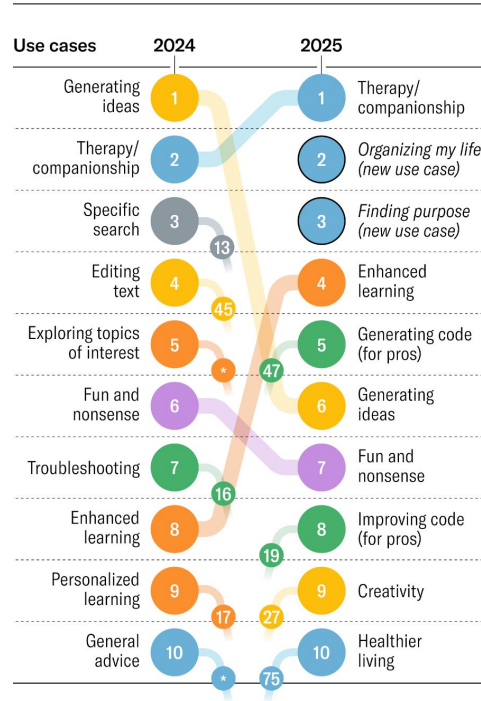
The hottest new programming language is English

12:14 PM · Jan 24, 2023 · **3.8M** Views

User interest has doubled in the last 12 months



Top 10 Gen AI Use Cases



Top 10 Gen AI Use Cases

The top 10 gen AI use cases in 2025 indicate a shift from technical to emotional applications, and in particular, growth in areas such as therapy, personal productivity, and personal development.

Themes

PERSONAL AND PROFESSIONAL SUPPORT	TECHNICAL ASSISTANCE AND TROUBLESHOOTING
CONTENT CREATION AND EDITING	CREATIVITY AND RECREATION
LEARNING AND EDUCATION	RESEARCH, ANALYSIS, AND DECISION-MAKING

Harvard Business Review

<https://hbr.org/2025/04/how-people-are-really-using-gen-ai-in-2025>

What exactly is an Large Language Models?

Simple Analogy:

Think of an LLM as a super-smart autocomplete on your phone. It's been fed massive amounts of text and code, and it's learned to predict what might come next.

More Technical:

A "large language model" is essentially a type of computer program adept at understanding and generating human language. It's trained on vast datasets consisting of text and code, enabling it to predict and generate coherent language based on the input it receives.

Translate English to Spanish:
What's the time?

What is the expected
weather in Puerto Rico in
December?

Summarize in 15 words
"During extra time, Messi
then scored again to give
Argentina a 3-2 lead.
However, Mbappé scored
another penalty to tie the
game 3-3 with only minutes
remaining, becoming the
second man to score a hat-
trick in a World Cup final.
Argentina then won the
ensuing penalty shoot-out
4-2 to win their third World
Cup."

Generate a tagline for a fruit
juice brand that focused on
sustainable and organic
farming.

LLM

Qué hora es?

In December, Puerto Rico
experiences pleasant
tropical weather with an
average high temperature of
84°F (29°C) and an average
low temperature of around
72°F (22°C).

Mbappé scores a hat-trick
but Argentina wins World
Cup in a penalty shoot-out.

Purely natural, sustainably
sourced. Taste the goodness
of organic fruits with every
sip.

Large Language Models

A Large Language Model (LLM) is a machine learning model designed to handle and generate human-like text based on vast amounts of training data and trained. Foundation Models

LLMs in Action: Cool Things They Can Do

Natural Language Understanding

LLMs excel at understanding and interpreting human language. They can comprehend context, infer meaning, and provide relevant responses to queries.

1

Language Translation

LLMs facilitate seamless language translation by converting text from one language to another while preserving context and meaning.

3

Creative Collaboration

LLMs can serve as collaborative partners in brainstorming sessions, suggesting ideas, refining concepts, and providing inspiration across diverse fields such as art, design, and storytelling

5

Text Generation

These models are capable of generating coherent and contextually relevant text. From writing articles to composing poetry, LLMs can produce content in various styles and genres.

2

Code Generation

With their understanding of programming languages, LLMs can assist developers by generating code snippets, automating repetitive tasks, and offering solutions to coding challenges.

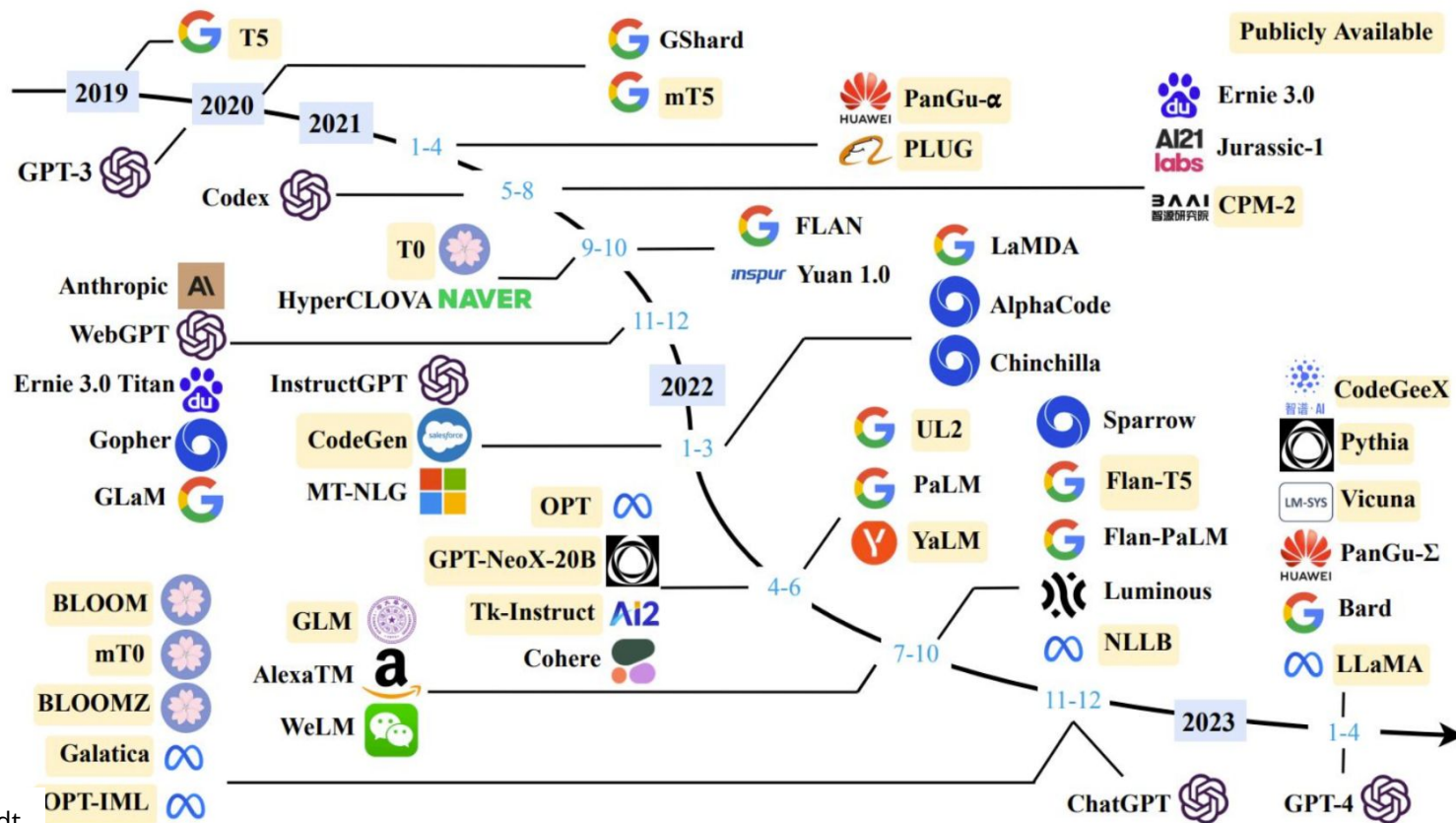
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Overall: Increase productivity!

LLMs can help boost productivity by automating tasks, generating content, and providing valuable insights.

6

LLMs since GPT-3



“Attention is All You Need.”

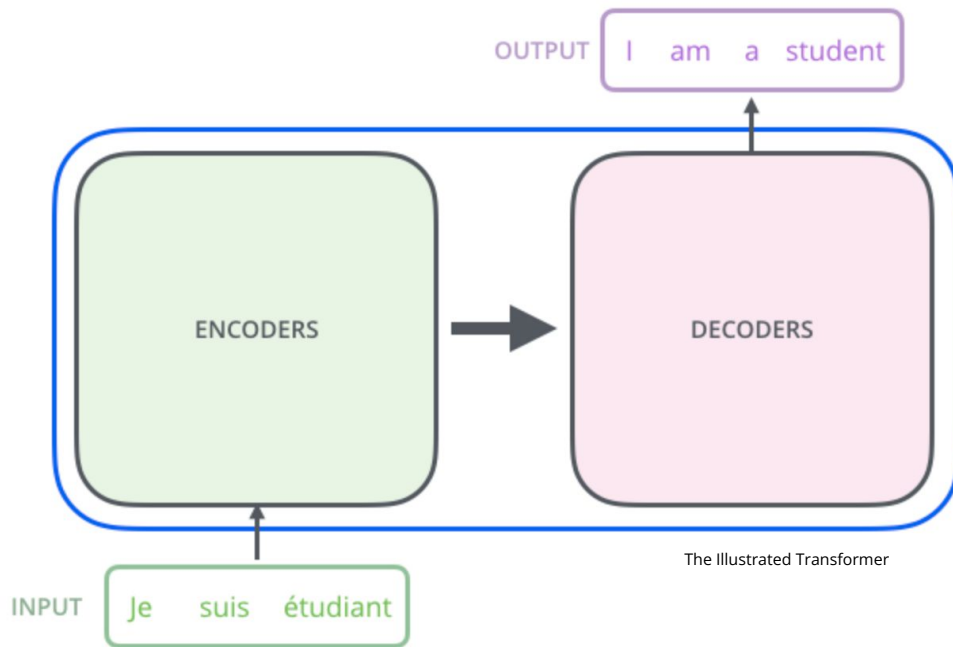
Transformers & Attention Mechanism:

Transformers - new architecture, which relies entirely on self-attention mechanisms. This enables the model to capture relationships in the data regardless of the distance between elements.

She poured water from the pitcher to **the cup** until **it was full**.

She poured water from **the pitcher** to the cup until **it was empty**.

Transformer architecture



Encoder (left): The encoder receives an input and builds a representation of it (its features). This means that the model is optimized to acquire understanding from the input. **The encoder reads and understands text.**

Decoder (right): The decoder uses the encoder's representation (features) along with other inputs to generate a target sequence. This means that the model is optimized for generating outputs. **The decoder writes text based on what it understands.**

Transformers models

Encoder-only models: Good for tasks that require understanding of the input, such as sentence classification and named entity recognition. (BERT (Google), RoBERTa (Meta), Sentence-BERT (SBERT))

Decoder-only models: Good for generative tasks such as text generation. (GPT-family, Llama-family, Google Gemini-family, DeepSeek, Mistral)

Encoder-decoder models or sequence-to-sequence models: Good for generative tasks that require an input, such as translation, summarization, question-answering (Google T5)

<https://poloclub.github.io/transformer-explainer/>

Encoder vs Decoder: What Are They Good At?

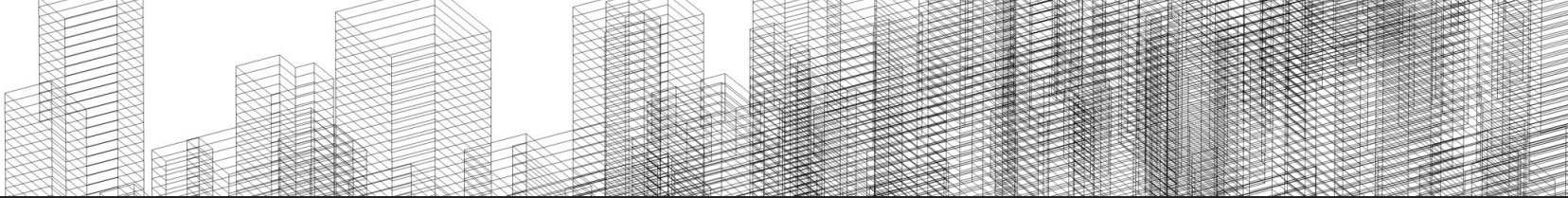
Encoders (Encode text):

- Semantic search
- Sentiment analysis
- Classification
- Named entity recognition
- Retrieval (for RAG systems)

Decoder-Only Models (LLMs)

Generate text → next-token prediction

- Text generation (chat, stories, explanations)
- Question answering
- Summarization
- Reasoning and chain-of-thought
- Code generation
- Dialogue and interaction
- RAG answer synthesis (not the retrieval part)



Sources to learn more about transformers

How do Transformers work?

<https://huggingface.co/learn/nlp-course/chapter1/4>

The Illustrated Transformer

<http://jalammar.github.io/illustrated-transformer/>

Kaggle & HuggingFace Integration

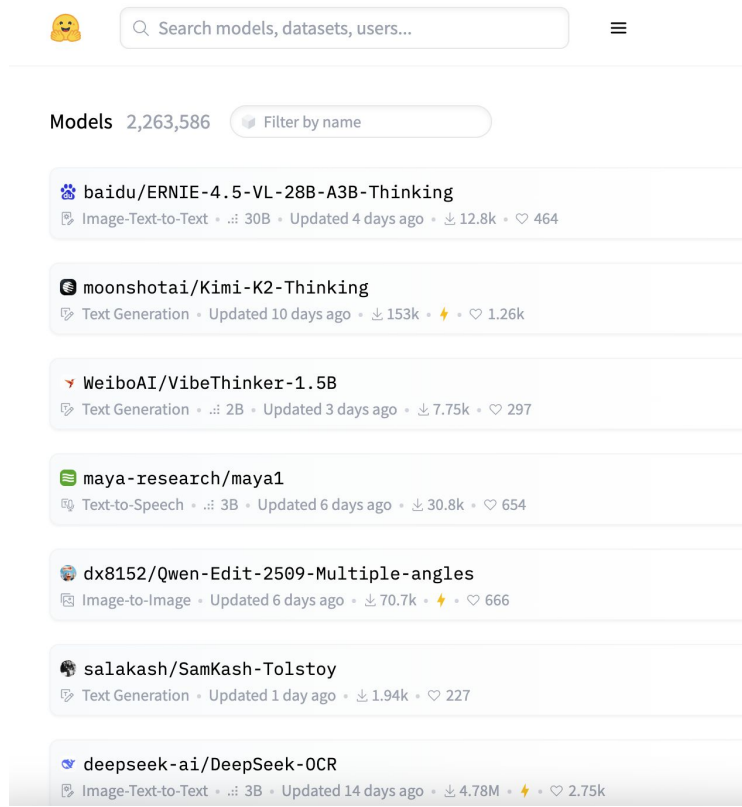
Hands-on. Hugging Face Transformers

<https://huggingface.co/>

Transfer Learning

- Identify the task type for your business problem
- Pick and test a pre-trained model
 - No need to prepare a large dataset
 - Less coding
- Optionally, fine-tune the model on your data
 - Much less data is required
 - No need to train for long periods of time

Which model to choose?



The screenshot shows the Hugging Face Models page. At the top, there is a search bar with the placeholder text "Search models, datasets, users..." and a hamburger menu icon to its right. Below the search bar, the text "Models 2,263,586" is displayed, followed by a "Filter by name" button. A list of models is shown below, each with its name, icon, and some statistics:

- baidu/ERNIE-4.5-VL-28B-A3B-Thinking**
Image-Text-to-Text • 30B • Updated 4 days ago • 12.8k • 464
- moonshotai/Kimi-K2-Thinking**
Text Generation • Updated 10 days ago • 153k • 1.26k
- WeiboAI/VibeThinker-1.5B**
Text Generation • 2B • Updated 3 days ago • 7.75k • 297
- maya-research/maya1**
Text-to-Speech • 3B • Updated 6 days ago • 30.8k • 654
- dx8152/Qwen-Edit-2509-Multiple-angles**
Image-to-Image • Updated 6 days ago • 70.7k • 666
- salakash/SamKash-Tolstoy**
Text Generation • Updated 1 day ago • 1.94k • 227
- deepseek-ai/DeepSeek-OCR**
Image-Text-to-Text • 3B • Updated 14 days ago • 4.78M • 2.75k

2 263 586 models!

<https://huggingface.co/models>

Large Language Models. Best practices

Achieve best possible performance

- Use best models (could be LLM or small model depends on the task)
- Use prompts with detailed task context, relevant information, instructions
- Experiment with prompt engineering techniques
- Experiment with few-shot examples that are 1) relevant to the test case, 2) diverse (if appropriate)
- Spend quality time optimizing a pipeline / "chain"

Optimize Cost

- Start optimizing for cost once you reach your expected performance
- Use techniques like Quantization, Distillation or smaller models trained on synthetic data